

STUDY QUESTIONS FOR CLASS DISCUSSION

RESEARCH METHODOLOGY

MODULE THREE

These questions are prepared to help students revise key areas in Research Methodology, including sources of research ideas, statement of the problem, literature review, research questions, research objectives, scope, significance, variables, hypothesis, and data analysis. The questions are based on the Module Three content provided.

A. Sources of Research Ideas

1. What is a research idea?
2. Mention five sources of research ideas.
3. Explain how previous studies can help a researcher identify a new research topic.
4. Using an example, explain how new technology such as social media can generate a research idea.
5. What is the difference between informal observation and systematic sources of research ideas?
6. Why should a beginner researcher consult experienced researchers before choosing a research topic?
7. Give one example of a research topic that can come from a social, academic, or business problem.

B. Statement of the Problem

8. What is a statement of the problem in a research proposal?
9. Why is the statement of the problem considered one of the most important parts of a research proposal?
10. Mention five characteristics of a good research problem.
11. What does it mean when we say that a research problem should address a gap in knowledge?
12. Why should a research problem be researchable?
13. Explain the three parts of a good statement of the problem:
 - a. The ideal situation
 - b. The reality/current situation
 - c. The consequences
14. Using any topic of your choice, write a three-paragraph statement of the problem.
15. Discuss three common mistakes students make when writing a statement of the problem.
16. Why should a statement of the problem be brief and focused?

C. Literature Review

17. What is a literature review?
18. Why is a literature review important in a research proposal?
19. How does a literature review help a researcher identify the research gap?
20. Mention and explain four types of literature review.
21. What is the difference between an argumentative review and an integrative review?
22. What is a systematic review?
23. Why is thematic organization commonly used in social science literature reviews?
24. Should a literature review only summarize previous studies? Explain your answer.
25. Why are citations important in a literature review?
26. What does it mean to “keep your own voice” when writing a literature review?
27. Using your own research topic, identify three themes you can use to organize your literature review.
28. Why should researchers use current and relevant sources?

D. Research Questions

29. What is a research question?
30. Why is a research question described as the heart of systematic investigation?
31. Mention five characteristics of a good research question.
32. What is the difference between a clear research question and a focused research question?
33. Why should a good research question not be answered by “yes” or “no” only?
34. Explain five steps involved in writing research questions.
35. Using your own research topic, formulate three research questions.
36. Discuss the difference between a research topic and a research question.
37. How can a research question influence research design and methodology?

E. Research Objectives

38. What are research objectives?
39. What is the difference between a general objective and specific objectives?
40. Why should research objectives begin with verbs such as *to examine*, *to assess*, *to determine*, or *to identify*?
41. Explain the importance of research objectives in a research proposal.

42. Explain the SMART characteristics of good research objectives.
43. Why must a research objective be measurable?
44. Using your own research topic, write one general objective and three specific objectives.
45. Discuss three common mistakes students make when writing research objectives.
46. Why should specific objectives correspond with research questions?

F. Scope of the Study

47. What is the scope of the study?
48. Why is the scope of the study important in a research proposal?
49. Mention the key elements that should be included in the scope of the study.
50. What is the difference between scope and limitations of the study?
51. Using your own topic, write the scope of the study by considering:
 - a. Population/sample
 - b. Geographical location
 - c. Time/duration
 - d. Topic or theory covered

G. Significance of the Study

52. What is the significance of the study?
53. Why should the significance of the study show who will benefit from the research?
54. Mention four groups that may benefit from a research study.
55. What is the difference between the general contribution and the specific contribution of a study?
56. Using your own topic, write the significance of the study for the following groups:
 - a. Students
 - b. Institution
 - c. Government
 - d. Future researchers

H. Research Variables

57. What is a variable in research?
58. Differentiate between an independent variable and a dependent variable.
59. Give an example of an independent variable and a dependent variable in a study on teaching methods and student performance.
60. What is an extraneous variable?
61. Why can extraneous variables affect research findings?

- 62. Mention five criteria for selecting research variables.
- 63. Why are variables important in the research process?
- 64. Using your own topic, identify the independent variable and dependent variable.

I. Research Hypothesis

- 65. What is a hypothesis?
- 66. What is the difference between a hypothesis and a prediction?
- 67. Why must a hypothesis be testable?
- 68. What is a null hypothesis?
- 69. What is an alternative hypothesis?
- 70. Give one example of a null hypothesis and one alternative hypothesis.
- 71. What is the difference between a directional hypothesis and a non-directional hypothesis?
- 72. What is the difference between a simple hypothesis and a complex hypothesis?
- 73. Mention five characteristics of a good hypothesis.
- 74. Why is a hypothesis important in scientific research?
- 75. Using your own topic, write one testable hypothesis.

J. Data Analysis

- 76. What is data analysis?
- 77. Why is data analysis important in research?
- 78. Mention five issues that researchers should consider during data analysis.
- 79. What is the difference between quantitative data analysis and qualitative data analysis?
- 80. What is descriptive analysis?
- 81. What is exploratory analysis?
- 82. What is diagnostic analysis?
- 83. What is predictive analysis?
- 84. What is prescriptive analysis?
- 85. Mention the five steps of data analysis.
- 86. Why is data cleaning important before data analysis?
- 87. What is the difference between analysis and interpretation?
- 88. Using an example, explain how data analysis can help a researcher answer a research question.

K. Applied Class Discussion Questions

- 89. Choose one research topic related to college students and write:
 - a. Statement of the problem
 - b. General objective

- c. Three specific objectives
 - d. Three research questions
90. For the topic “**The Influence of Social Media Use on Academic Performance among University Students,**” identify:
 - a. Independent variable
 - b. Dependent variable
 - c. Extraneous variables
 - d. Possible hypothesis
 91. For the topic “**Challenges Facing Small Entrepreneurs in Accessing Loans,**” write a statement of the problem using the three parts: ideal situation, reality, and consequences.
 92. Prepare a literature review outline for any topic in business, education, accounting, procurement, or entrepreneurship.
 93. Why should a student not start writing the methodology before having clear objectives and research questions?
 94. Discuss with examples how research objectives guide data collection and data analysis.
 95. Explain why a good research proposal must show alignment between the title, problem statement, objectives, research questions, methodology, and data analysis.

L. Revision Questions for Test Preparation

96. What is the difference between a research aim and research objectives?
97. Why should a research problem be significant, ethical, and researchable?
98. What are the main purposes of a literature review?
99. Why should research questions be clear, focused, and complex?
100. What is the importance of SMART objectives in research?
101. How do variables help a researcher measure concepts?
102. Why is a null hypothesis written in negative form?
103. Why should researchers avoid biased interpretation during data analysis?
104. What is the relationship between data collection methods and data analysis methods?
105. Why are reliability and validity important in research?